MATHEMATICS / INDEPENDENT STUDY

MM8036

Type Theory

Exercises and mock assessments

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 1 · 24 October 2026

Contents

1 Unit 1 · Binding and substitution 3
2 Unit 2 · Reduction and confluence 5
3 Unit 3 · Simple types and judgments 7
4 Unit 4 · Normalization and constructive logic 9
5 Unit 5 · Dependent types 11
6 Unit 6 · Induction, identity, and universes 13
7 Unit 7 · Substitutions and semantics 15
8 Unit 8 · Categorical models and proof assistants 17
9 Mock assessment 1 19
10 Mock assessment 2 20

How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

1 Unit 1 · Binding and substitution

MM8036-U1.01 · E · 15 min
Learning objective: U1.1 Parsing
Fully parenthesize $\lambda x.xyz$ and $ab(cd)$.

MM8036-U1.02 · E · 15 min
Learning objective: U1.1 Free variables
Compute $\mathrm{FV}(\lambda x.(xy)(\lambda y.yz))$.

MM8036-U1.03 · E · 15 min
Learning objective: U1.1 Alpha equivalence
Which of $\lambda x.\lambda y.x$, $\lambda a.\lambda b.a$, and $\lambda a.\lambda b.b$ are alpha equivalent?

MM8036-U1.04 · E · 15 min
Learning objective: U1.2 Substitution
Compute $(\lambda y.xy)[x := yz]$.

MM8036-U1.05 · C · 30 min
Learning objective: U1.2 Substitution under shadowing
Compute $(\lambda x.(\lambda y.xy)z)[y := x]$ and explain why the replacement has no effect.

MM8036-U1.06 · C · 30 min
Learning objective: U1.2 Exact free-variable formula
Prove: if $x \in \mathrm{FV}(t)$, then $\mathrm{FV}(t[x := u]) = (\mathrm{FV}(t) \setminus \{x\}) \cup \mathrm{FV}(u)$.

MM8036-U1.07 · C · 30 min
Learning objective: U1.2 Closed substitutions

If t has free variables among $x_1, \ldots, x_n$ and each u_i is closed, prove simultaneous substitution of u_i for x_i closes t.

MM8036-U1.08 · C · 30 min
Learning objective: U1.2 Composition law

Verify the substitution-composition law for $t = xy$, $u = yz$, $v = w$, substituting first x then y.

MM8036-U1.09 · C · 30 min
Learning objective: U1.1 Structural induction

Let size count one for each variable, application, and abstraction node. Prove $|t[x := u]| = |t| + k(|u| - 1)$, where k is the number of free occurrences of x in t.

MM8036-U1.10 · A · 50 min
Learning objective: U1.2 Necessity of hypotheses

Construct a counterexample to the substitution-composition equation when $x \in \mathrm{FV}(v)$, even though $x \neq y$.

MM8036-U1.11 · A · 50 min
Learning objective: U1.2 Commuting substitutions

Prove that substitutions for distinct x, y commute if $x \notin \mathrm{FV}(v)$ and $y \notin \mathrm{FV}(u)$: $(t[x := u])[y := v] = (t[y := v])[x := u]$ up to alpha equivalence. Explain why both side conditions are needed.

MM8036-U1.12 · A · 50 min
Learning objective: U1.1 Binding invariants

Use zero-based de Bruijn indices to encode the closed terms $\lambda x.\lambda y.x(yx)$ and $\lambda a.\lambda b.a(ba)$. Explain why this representation detects their alpha equivalence, and why a rule for free variables is still needed for open terms.

2 Unit 2 · Reduction and confluence

MM8036-U2.01 · E · 15 min
Learning objective: U2.1 Beta reduction
Reduce $(\lambda x.x)(\lambda y.y)$ to normal form.

MM8036-U2.02 · E · 15 min
Learning objective: U2.1 Capture avoidance
Reduce $(\lambda x.\lambda y.x)y$.

MM8036-U2.03 · E · 15 min
Learning objective: U2.1 Divergence
Let $\Delta = \lambda x.xx$ and $\Omega = \Delta\Delta$. Show an infinite beta sequence from Ω.

MM8036-U2.04 · E · 15 min
Learning objective: U2.1 Reduction strategies
Compare outermost reduction and call-by-value on $(\lambda x.\lambda z.z)\Omega$.

MM8036-U2.05 · C · 30 min
Learning objective: U2.2 Joining a fork
For $(\lambda x.xx)((\lambda y.y)z)$, contract either the outer redex or the argument redex first and exhibit a common reduct.

MM8036-U2.06 · C · 30 min
Learning objective: U2.2 Normal-form uniqueness
Derive uniqueness of beta normal forms up to alpha equivalence from confluence.

MM8036-U2.07 · C · 30 min
Learning objective: U2.2 Conversion

Show that beta-convertible normal terms are alpha equivalent. You may use Church–Rosser for conversion.

MM8036-U2.08 · C · 30 min
Learning objective: U2.1 Weak versus strong normalization

Prove that $(\lambda x.\lambda z.z)\Omega$ is weakly normalizing but not strongly normalizing under full beta reduction.

MM8036-U2.09 · C · 30 min
Learning objective: U2.2 Parallel reduction

Compute the complete development of $(\lambda x.xx)((\lambda y.y)z)$. Explain why it differs from contracting just one redex.

MM8036-U2.10 · A · 50 min
Learning objective: U2.2 Confluence versus termination

Give a finite abstract reduction system that is confluent but has an infinite reduction sequence, and prove both claims.

MM8036-U2.11 · A · 50 min
Learning objective: U2.2 Substitution and reduction

Prove that if $u \to u'$, then $t[x := u] \xrightarrow{*} t[x := u']$. Explain why one step is generally insufficient.

MM8036-U2.12 · A · 50 min
Learning objective: U2.2 Distinguishing convertibility

Prove the closed terms $\lambda x.\lambda y.x$ and $\lambda x.\lambda y.y$ are not beta convertible. Why is their different behavior on some arguments not needed for this proof?

3 Unit 3 · Simple types and judgments

MM8036-U3.01 · E · 15 min
Learning objective: U3.1 Typing judgments
Give a derivation of $\lambda x.x : A \to A$ in the empty context.

MM8036-U3.02 · E · 15 min
Learning objective: U3.1 Application
In context $f : A \to B, x : A$, derive $fx : B$.

MM8036-U3.03 · E · 15 min
Learning objective: U3.1 Arrow association
Give a type and derivation for $\lambda x.\lambda y.x$.

MM8036-U3.04 · E · 15 min
Learning objective: U3.2 Substitution
Given $x : A \vdash x : A$ and $\vdash u : A$, calculate the conclusion of substitution.

MM8036-U3.05 · C · 30 min
Learning objective: U3.1 Composition
Derive the type of $\lambda f.\lambda g.\lambda x.f(gx)$ from chosen types $f : B \to C, g : A \to B$.

MM8036-U3.06 · C · 30 min
Learning objective: U3.2 Weakening
Prove admissibility of adding a fresh unused variable $y : B$ to a simply typed context.

MM8036-U3.07 · C · 30 min
Learning objective: U3.2 Preservation

Prove preservation for a root redex $(\lambda x.t)u \to t[x := u]$ using inversion and substitution.

MM8036-U3.08 · C · 30 min
Learning objective: U3.2 Open terms and progress

Give a well-typed open term that is neither a call-by-value value nor able to step, under the lecture convention that values are abstractions and the inert base constant.

MM8036-U3.09 · C · 30 min
Learning objective: U3.1 Untypability of self-application

Prove $\lambda x.xx$ is untypable in finite simple types.

MM8036-U3.10 · A · 50 min
Learning objective: U3.2 Failure of subject expansion

Construct a beta step from an untypable term to a typable closed term. Prove both typing assertions.

MM8036-U3.11 · A · 50 min
Learning objective: U3.2 Substitution proof

Prove the application and abstraction cases of the substitution lemma, including why a binder may need renaming.

MM8036-U3.12 · A · 50 min
Learning objective: U3.1 Typing restricts syntax

Prove that a closed beta-normal arrow-only term of base type o, in the language with exactly one inert constant $c : o$, must be c.

4 Unit 4 · Normalization and constructive logic

MM8036-U4.01 · E · 15 min
Learning objective: U4.2 Curry–Howard
Write a proof term for $A \to B \to A$.

MM8036-U4.02 · E · 15 min
Learning objective: U4.2 Conjunction
Construct $(A \times B) \to A$ and $(A \times B) \to B$.

MM8036-U4.03 · E · 15 min
Learning objective: U4.2 Disjunction
Construct $A \to A + B$ and explain what data the output contains.

MM8036-U4.04 · E · 15 min
Learning objective: U4.1 Normalization vocabulary
State weak and strong normalization and their implication relation.

MM8036-U4.05 · C · 30 min
Learning objective: U4.2 Negation
Construct $(A \to B) \to ((B \to 0) \to (A \to 0))$.

MM8036-U4.06 · C · 30 min
Learning objective: U4.2 Sum elimination
Construct $((A \to C) \times (B \to C)) \to ((A + B) \to C)$ and state its two computation cases.

MM8036-U4.07 · C · 30 min
Learning objective: U4.1 Candidate closure
Explain why every variable belongs to every reducibility candidate defined in Lecture 7.

MM8036-U4.08 · C · 30 min
Learning objective: U4.1 Why strengthen induction
Show that strong normalization of s and t does not imply strong normalization of st.

MM8036-U4.09 · C · 30 min
Learning objective: U4.2 Double negation
Construct $A \to ((A \to 0) \to 0)$ and explain why reversing the arrow requires a different argument.

MM8036-U4.10 · A · 50 min
Learning objective: U4.1 Canonicity consequence
Using normalization, preservation, and U3.12, prove every closed term of type o reduces to c in the arrow-only calculus with sole constant $c : o$.

MM8036-U4.11 · A · 50 min
Learning objective: U4.2 Constructive equivalence
Construct maps in both directions between $(A + B) \to C$ and $(A \to C) \times (B \to C)$. Show the composites agree pointwise on constructor inputs.

MM8036-U4.12 · A · 50 min
Learning objective: U4.1 Normalization proof dependencies
Reconstruct the abstraction case of the fundamental reducibility lemma. Identify exactly where the controlled beta-expansion lemma is used.

5 Unit 5 · Dependent types

MM8036-U5.01 · E · 15 min
Learning objective: U5.1 Pi formation
State the data needed to form $\Pi_{x:A} B(x)$ in context Γ.

MM8036-U5.02 · E · 15 min
Learning objective: U5.1 Dependent application
If $f : \Pi_{x:A} B(x)$ and $a : A$, what is the type of $f a$?

MM8036-U5.03 · E · 15 min
Learning objective: U5.2 Sigma introduction
What must be supplied to construct a term of $\Sigma_{x:A} B(x)$?

MM8036-U5.04 · E · 15 min
Learning objective: U5.2 Sigma projections
State the types and computation rules for the projections of $p : \Sigma_{x:A} B(x)$.

MM8036-U5.05 · C · 30 min
Learning objective: U5.1 Dependent identity
Derive the type of $\lambda A.\lambda x.x$ with $A : U_0$. Determine the level of its type.

MM8036-U5.06 · C · 30 min
Learning objective: U5.2 Dependent pair map
Given $f : A \to A'$ and $g : \Pi_{x:A}(B(x) \to B'(fx))$, construct a map $\Sigma_{x:A} B(x) \to \Sigma_{y:A'} B'(y)$.

MM8036-U5.07 · C · 30 min
Learning objective: U5.1 Dependent composition
Given $f : \Pi_{x:A} B(x)$ and $g : \Pi_{x:A}(B(x) \to C(x))$, derive a term of $\Pi_{x:A} C(x)$.

MM8036-U5.08 · C · 30 min
Learning objective: U5.2 Existential evidence
Explain the computational difference between $\Sigma_{n:\mathrm{Nat}} P(n)$ and $\Pi_{n:\mathrm{Nat}} P(n)$, and give their respective elimination uses.

MM8036-U5.09 · C · 30 min
Learning objective: U5.2 Constant families
Show that Sigma and Pi over constant families reproduce ordinary products and arrows.

MM8036-U5.10 · A · 50 min
Learning objective: U5.2 Constructive choice
Construct a function from $\Pi_{x:A} \Sigma_{y:B(x)} R(x, y)$ to $\Sigma_{f:\Pi_{x:A} B(x)} \Pi_{x:A} R(x, fx)$. Explain why this does not introduce classical choice.

MM8036-U5.11 · A · 50 min
Learning objective: U5.2 Dependent currying
Construct maps between $\Pi_{p:\Sigma_{x:A} B(x)} C(\pi_1(p), \pi_2(p))$ and $\Pi_{x:A} \Pi_{y:B(x)} C(x, y)$. Verify their computation on pair inputs.

MM8036-U5.12 · A · 50 min
Learning objective: U5.1 Context dependency
Explain why exchanging $x : A, y : B(x)$ into $y : B(x), x : A$ is generally invalid, although weakening by a fresh declaration is often admissible. Give a concrete family example.

6 Unit 6 · Induction, identity, and universes

MM8036-U6.01 · E · 15 min
Learning objective: U6.1 Addition computation
With addition recursive in its first argument, compute $\mathrm{add}(\mathrm{suc}(\mathrm{zero}), n)$.

MM8036-U6.02 · E · 15 min
Learning objective: U6.1 Induction motive
State the motive for proving right-zero addition.

MM8036-U6.03 · E · 15 min
Learning objective: U6.2 Transport typing
For $p : \mathrm{Id}_{A(a,b)}$ and $u : B(a)$, state the type of $\mathrm{transport}_{B(p,u)}$ and its reflexivity computation.

MM8036-U6.04 · E · 15 min
Learning objective: U6.2 Universe levels
If $A : U_1$ and $B(x) : U_0$, find a universe containing $\Pi_{x:A} B(x)$ and one containing $\mathrm{Id}_{A(a,b)}$.

MM8036-U6.05 · C · 30 min
Learning objective: U6.1 Associativity of addition
Prove addition associative by induction on its first argument.

MM8036-U6.06 · C · 30 min
Learning objective: U6.1 Vector append
Describe and type both clauses of append on length-indexed vectors.

MM8036-U6.07 · C · 30 min
Learning objective: U6.2 Symmetry by J
Derive symmetry of identity with an explicit motive and reflexive case.

MM8036-U6.08 · C · 30 min
Learning objective: U6.2 Congruence
For $f : A \to D$, derive $\mathrm{Id}_{A(a,b)} \to \mathrm{Id}_{D(f(a),f(b))}$. Why is this an implication about identity proofs rather than an equality-reflection rule?

MM8036-U6.09 · C · 30 min
Learning objective: U6.1 Safe head
Explain why a head function $\mathrm{Vec}(A, \mathrm{suc}(n)) \to A$ needs no nil branch. Can the same total signature be given for all lengths without extra data?

MM8036-U6.10 · A · 50 min
Learning objective: U6.2 Transport composition
For $p : \mathrm{Id}_{A(a,b)}$ and $q : \mathrm{Id}_{A(b,c)}$, prove transport along their composite agrees propositionally with successive transport on $u : B(a)$.

MM8036-U6.11 · A · 50 min
Learning objective: U6.2 Judgmental and propositional equality
Compare the left-zero and right-zero laws for addition. Explain why a proof of right-zero does not make its open expression reduce by definition.

MM8036-U6.12 · A · 50 min
Learning objective: U6.2 Diagnosing misuse of path induction
A student argues: “Apply J to $p : \mathrm{Id}_{A(a,a)}$, so every such p equals reflexivity.” Identify the missing motive issue; explain what J actually supplies.

7 Unit 7 · Substitutions and semantics

MM8036-U7.01 · E · 15 min
Learning objective: U7.1 Context substitution
Describe a substitution from Δ to the context $x : A, y : B(x)$.

MM8036-U7.02 · E · 15 min
Learning objective: U7.1 Identity substitution
Write the identity substitution on $x : A, y : B(x)$.

MM8036-U7.03 · E · 15 min
Learning objective: U7.1 Composition order
For $\sigma : \Delta \to \Gamma$ and $\tau : \Theta \to \Delta$, state the source and target of $\sigma \circ \tau$ and the substitution equation for a term t over Γ.

MM8036-U7.04 · E · 15 min
Learning objective: U7.2 Meaning explanations
Contrast a canonical natural-number term with a neutral natural-number term.

MM8036-U7.05 · C · 30 min
Learning objective: U7.1 Associativity on components
Prove associativity of context-substitution composition from the term substitution law.

MM8036-U7.06 · C · 30 min
Learning objective: U7.1 Projection and pairing
For context extension $\Gamma \cdot A$, verify $p \circ \langle \sigma, t \rangle = \sigma$ and $q[\langle \sigma, t \rangle] = t$ using tuple substitutions.

MM8036-U7.07 · C · 30 min
Learning objective: U7.2 Families of sets
In the set interpretation, describe the meaning of a dependent sum for $A = \{0, 1\}$, $B(0) = \{a\}$, $B(1) = \{b, c\}$. List its elements.

MM8036-U7.08 · C · 30 min
Learning objective: U7.2 Dependent products of sets
For the same two fibers as U7.07, list all dependent functions and count them.

MM8036-U7.09 · C · 30 min
Learning objective: U7.2 Semantic substitution
Explain why interpreting $t[\sigma]$ agrees with evaluating the interpretation of t in the environment determined by σ. Prove the variable and application cases.

MM8036-U7.10 · A · 50 min
Learning objective: U7.2 Soundness is not completeness
The set model interprets every identity proof between fixed endpoints, when one exists, by the same element. Explain why soundness alone does not prove syntactic UIP.

MM8036-U7.11 · A · 50 min
Learning objective: U7.1 Explicit substitution design
Why does the equation $(t[\sigma])[\tau] = t[\sigma \circ \tau]$ not by itself prove that an explicit-substitution rewrite system terminates? Construct a simple warning using orientations.

MM8036-U7.12 · A · 50 min
Learning objective: U7.2 Reindexing a family
Let $A = \{0, 1\}$ with fibers as in U7.07, and let $\sigma : \{r, s, t\} \to A$ send r, s to zero and t to one. Describe the reindexed family, its total space, and its number of sections.

8 Unit 8 · Categorical models and proof assistants

MM8036-U8.01 · E · 15 min
Learning objective: U8.1 Context morphisms
Explain how contexts and substitutions form a category, naming identities and composition.

MM8036-U8.02 · E · 15 min
Learning objective: U8.1 Terminal context
Why is the empty context terminal in the substitution category?

MM8036-U8.03 · E · 15 min
Learning objective: U8.1 Evaluation
In sets, write evaluation $B^A \times A \to B$ and the curry of $h : X \times A \to B$.

MM8036-U8.04 · E · 15 min
Learning objective: U8.2 Checker contract
What does successful checking of a lab's theorem establish, and what does it leave to the author?

MM8036-U8.05 · C · 30 min
Learning objective: U8.1 Exponential uniqueness
In sets, prove the curry in U8.03 is the unique $k : X \to B^A$ satisfying $\mathrm{eval} \circ (k \times 1_A) = h$.

MM8036-U8.06 · C · 30 min
Learning objective: U8.1 Beta soundness
Explain categorically why interpreting an application of an abstraction satisfies the beta equation.

MM8036-U8.07 · C · 30 min
Learning objective: U8.1 Comprehension in sets

For a set Γ and family $A(\gamma)$, construct the context extension and prove its pairing property.

MM8036-U8.08 · C · 30 min
Learning objective: U8.2 Debugging an induction

An attempted proof of right-zero addition uses reflexivity for the variable case. Explain the failure and repair it using the lab definitions.

MM8036-U8.09 · C · 30 min
Learning objective: U8.2 Transport in code

Describe how to cast a vector of length m to one of length n given $p : \mathrm{Id}_{\mathrm{Nat}(m,n)}$. State its reflexive computation.

MM8036-U8.10 · A · 50 min
Learning objective: U8.1 CCC and dependent models

Explain why a cartesian closed category by itself supplies a model of simple arrows but does not automatically provide all the dependent-type structure used here.

MM8036-U8.11 · A · 50 min
Learning objective: U8.2 Constructive proof audit

Construct a program of type $(A + (A \to 0)) \to (((A \to 0) \to 0) \to A)$. Explain both branches, why this does not establish unrestricted double-negation elimination, and why replacing the missing decision input by a postulate would violate the completed-lab contract.

MM8036-U8.12 · A · 50 min
Learning objective: U8.1 Synthesis of substitution and semantics

In the families-of-sets model, prove that reindexing a term along $\sigma : \Delta \to \Gamma$ and then $\tau : \Theta \to \Delta$ equals reindexing along $\sigma \circ \tau$. Relate this to the syntactic explicit-substitution equation.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM8036-M1.01 · E · 15 min
Learning objective: U1–U2 Binding and reduction
Compute $\mathrm{FV}((\lambda x.\lambda y.xy)z)$ and reduce the term to normal form.

MM8036-M1.02 · E · 15 min
Learning objective: U3–U4 Typing and logic
Derive the type $(A \to B) \to A \to B$ and interpret the term as a logical proof.

MM8036-M1.03 · C · 30 min
Learning objective: U2–U4 Normalization
Give a weakly but not strongly normalizing term, then explain why it cannot be typable in the arrow-only core.

MM8036-M1.04 · C · 30 min
Learning objective: U5–U6 Dependent data
Define vector append on paper and justify its result index in both cases.

MM8036-M1.05 · A · 50 min
Learning objective: U3–U4 Metatheory synthesis
In the arrow-only language with sole constant $c : o$, show every closed term of type $o \to o$, applied to c, reduces to c. Explain what this does and does not say about equality of functions.

MM8036-M1.06 · A · 50 min
Learning objective: U5–U8 Proof and model
Construct the dependent-pair map of U5.06 and explain how the same formula acts on total spaces in the set model. Prove it respects the projection to the new base.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM8036-M2.01 · E · 15 min
Learning objective: U1–U3 Substitution
Reduce $(\lambda x.\lambda z.xz)y$ and give its type under $y : A \to B$.

MM8036-M2.02 · E · 15 min
Learning objective: U5–U6 Identity
Give the type and defining reflexive case of transport in a family $P : A \to U_0$.

MM8036-M2.03 · C · 30 min
Learning objective: U4–U5 Constructive evidence
Construct a term taking $\Sigma_{x:A}(B(x) \times C(x))$ to $(\Sigma_{x:A}B(x)) \times (\Sigma_{x:A}C(x))$. Explain why the reverse direction is generally unavailable.

MM8036-M2.04 · C · 30 min
Learning objective: U6 Induction
Prove that mapping the identity function over a vector returns the original vector, using propositional equality.

MM8036-M2.05 · A · 50 min
Learning objective: U2–U4 Confluence and typing
Suppose a closed simply typed term reduces along two paths to u, v. Prove u, v have a common normal form of the original type.

MM8036-M2.06 · A · 50 min
Learning objective: U6–U8 Equality audit
Explain why the following inference is invalid: “The set model identifies all proofs of equality; therefore the paper theory may use K, and the labs may omit `--without-K`.” Give a correct relationship among syntax, semantics, and the lab options.